Many Daughters
Pre-analysis plan Champeaux & Gazeaud

Overall identification strategy

We are testing causal hypotheses and therefore need a proper identification strategy. OLS estimates are likely biased due to potential unobserved confounders, such as parental preferences or values that simultaneously influence the likelihood of having daughters and attitudes toward gender equality. To estimate the causal effect of having daughters on support for gender equality we suggest using an IV strategy. Specifically, we aim to instrument the (endogenous) indicator for having at least one daughter with a binary variable equal to one if the respondent's first-born child is a girl. The core identifying assumption is that the sex of the first-born child is as good as random (a reasonable assumption in the absence of selective abortions) and affects the outcome only through its impact on whether the respondent has a daughter.

This exclusion restriction may be violated if parents adjust their fertility behavior based on the sex of their first-born children. For example, in the presence of a son preference, parents whose first child is a girl may be more likely to have additional children in an attempt to have a son. As a result, the instrument could influence outcomes not only through the likelihood of having a daughter but also via total family size. To mitigate this concern, we will include the number of children as a control variable.

To improve the precision of our estimates, we will also include a set of pre-determined respondent characteristics that are unlikely to be affected by the instrument. These include age, age squared, age at first birth, age at first birth squared. Including these variables helps us to reduce residual variance in the outcome without threatening the exclusion restriction, thereby increasing statistical power.

We pool observations from all SOEP waves in which the outcome variable is available and cluster standard errors at the individual (resp. household) level to account for within-person (resp. within-household) correlation across survey waves. We include survey round fixed effects to account for time-specific shocks and general trends in the outcome. We do *not* include individual (resp. household) fixed effects because it would restrict identification to short-term within-person (resp. household) changes, whereas we aim to capture both the direct and indirect effects of having daughters, including those that develop over longer-term interactions as daughters reach adulthood.

We do not correct for outliers. For each control variable, we replace missing values with -99 and include in our regressions dummy variables to indicate observations with originally missing values. This approach allows us to retain all observations even when some control variables have missing data.

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes).

Variables

We use the following questions to construct an index measuring support for gender equality. These questions use scales where the highest score indicates full agreement with the

statement and the lowest score indicates full disagreement. Variables plj0623, plj0624, and plj0625 are reverse-coded so that higher values indicate greater support for gender equality.

Table 1. Questions used to construct the index of support for gender equality

Name	Label	Type	Year
plj0620	Women have the same rights as men.	Scale 1-10[1]	2016
plj0621	Having a job is the best way for a woman to be independent.	Likert 1-7[2]	2016
plj0622	Even a married woman should have a paid job so that she can be financially independent.	Likert 1-7	2016
plj0623	If a woman earns more money than her partner, this inevitably leads to problems.	Likert 1-7	2016
plj0624	For parents, vocational training or higher education for their sons should be more important than vocational training or higher education for their daughters.	Likert 1-7	2016
plj0625	At home, the husband should have the final say	Likert 1-7	2016

To construct the index, we first recode each variable into a binary indicator equal to 1 if the response indicates support for gender-egalitarian values. For plj0620 (scale: 1-10), we consider responses from 6 to 10 as indicative of support for gender equality. For the other variables (scale: 1-7), we consider responses from 5 to 7 as indicative of support for gender equality. These cutoffs are selected to capture only responses expressing support for gender equality. The final index is the sum of these binary indicators, which ranges from 0 to 6.

Inclusion/exclusion criteria

We restrict the sample to adults under the age of 65 with at least one child in the 2016 wave of the SOEP—the only round in which our outcome variables are available.

Exact analysis

We will estimate the following model using two-stage least squares (2SLS):

$$Y_i = \pi_0 + \pi_1 \cdot HasDaughter_i + X_i \cdot \pi + u_i$$

Where Y_i is the index of support for gender equality for respondent *i*, as defined above. $HasDaughter_i$ is the dummy equal to one if the respondent has at least one daughter. Since it is endogenous, we instrument it using $FirstBornGirl_i$, a dummy equal to one if the first-born child of respondent is a girl. X_i is a vector of control variables including the respondent's age, age squared, age at first birth, age at first birth squared, and the total number of children, as well as dummy variables for each of these controls indicating whether the original value was missing and imputed with -99. We use heteroskedasticity-robust standard errors.

[1] Respondents answer using a scale from 0 to 10, where 0 means "Should definitely not happen in a democracy" and 10 means "Should definitely happen in a democracy."
[2] For this scale (and the following 1-7 Likert type questions), respondents were asked to answer using a scale where 1 means "I totally disagree" and 7 means "I totally agree".

Interpretation of results

The parameter of interest is π_1, which captures the causal effect of having at least one daughter on support for gender equality. We consider that the evidence is consistent with H1 if the estimate is positive and the two-sided p-value is below 0.05.

H2. Having daughters leads to more left-leaning political preferences.

Variables

We use the following question as our indicator of left-leaning political preferences. Enumerators ask the respondent to classify him/herself on a scale from 1 to 10, with 1 corresponding to the extreme left, and 10 to the extreme right. We reverse-code the variable so that higher values indicate more left-leaning political preferences.

Table 2. Questions used to measure left-leaning political preferences

Name	Label	Type	Year
plh0004	In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?	Scale 1-10	2005, 2009, 2014, 2019

Inclusion/exclusion criteria

Similar to H1, we restrict the sample to adults under the age of 65 with at least one child. We use the 2005, 2009, 2014 and 2016 waves of the SOEP—all the rounds in which our outcome variable is available.

Exact analysis

We will estimate the following model using two-stage least squares (2SLS):

$$Y_{it} = \pi_0 + \pi_1 \cdot HasDaughter_{it} + X_{it} \cdot \pi + \gamma_t + u_{it}$$

Where Y_{it} is a dummy variable indicating whether respondent *i* in wave *t* has left-leaning political preferences, as defined above. $HasDaughter_i$ is the dummy equal to one if the respondent has at least one daughter. Since it is endogenous, we instrument it using $FirstBornGirl_i$, a dummy equal to one if the first-born child of respondent is a girl. X_i is a vector of control variables including the respondent's age, age squared, age at first birth, age at first birth squared, and total number of children, as well as dummy variables for each of these controls indicating whether the original value was missing and imputed with -99. γ_t are survey round fixed effects. Heteroskedasticity-robust standard errors are clustered at the individual level to account for within-person correlation across survey waves.

Interpretation of results

The parameter of interest is π_1, which captures the causal effect of having at least one daughter on the likelihood the respondents have left-leaning political preferences. We

consider that the evidence is consistent with H2 if the estimate is positive and the two-sided p-value is below 0.05.

H3. Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Variables

We use the following questions to construct an index of gender equality in the household. For these variables, enumerators asked respondents to report the number of hours spent on household chores and childcare on weekdays, on Saturdays, and on Sundays.

Table 3. Questions used to measure gender equality in the household

Name	Label	Type	Year
pli0043_h	Time use for different activities (weekdays): Chores	0-24 (hours)	All
pli0044_h	Time use for different activities (weekdays): Childcare	0-24 (hours)	All
pli0012_h	Time use for different activities (Saturdays): Chores	0-12 (hours)	2001; 2003; 2005; 2007; 2009; 2011; 2012; 2013; 2015; 2017; 2019;
pli0019_h	Time use for different activities (Saturdays): Childcare	0-24 (hours)	Same than pli0012_h
pli0016_h	Time use for different activities (Sundays): Chores	0-12 (hours)	Same than pli0012_h
pli0022_h	Time use for different activities (Sundays): Childcare	0-24 (hours)	Same than pli0012_h

We compute at the household level the father's share of domestic work (childcare and chores) relative to the total performed by the couple. To do so, we focus on survey waves that include time use data for both weekdays and weekends (i.e., the years 2001, 2003, 2005, 2007, 2009, 2011, 2012, 2013, 2015, 2017, and 2019). We then compute the total hours each partner spends on domestic work. Finally, we derive the share performed by men and and retain one observation per household per survey round.

Inclusion/exclusion criteria

We restrict the sample to nuclear households in which adult men are under the age of 65, have at least one child, and live with a spouse or a partner. Nuclear households are defined

as households containing no more than one couple. We use the 2001, 2003, 2005, 2007, 2009, 2011, 2012, 2013, 2015, 2017, 2019 waves of the SOEP—all the rounds in which our outcome variable is available.

Exact analysis

We will estimate the following model using two-stage least squares (2SLS) :

$$Y_{it} = \pi_0 + \pi_1 \cdot HasDaughter_{it} + X_{it} \cdot \pi + \gamma_t + u_{it}$$

Where Y_{it} is the share of domestic work performed by the father in household *i* in wave *t*, as defined above. $HasDaughter_i$ is the dummy equal to one if the household has at least one daughter. Since it is endogenous, we instrument it using $FirstBornGirl_i$, a dummy equal to one if the first-born child of the household is a girl. X_i is a vector of control variables including the household's age, age squared, age at first birth, age at first birth squared, age of the partner, age of the partner squared, and total number of children, as well as dummy variables for each of these controls indicating whether the original value was missing and imputed with -99. γ_t are survey round fixed effects. Heteroskedasticity-robust standard errors are clustered at the household level to account for within-household correlation across survey waves.

Interpretation of results

The parameter of interest is π_1, which captures the causal effect of having at least one daughter on the share of housework performed by the father. We consider that the evidence is consistent with H3 if the estimate is positive and the two-sided p-value is below 0.05.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Variables

Same as H1.

Inclusion/exclusion criteria

Same as H1.

Exact analysis

We will estimate the following model using two-stage least squares (2SLS):

$$Y_i = \pi_0 + \pi_1 \cdot HasDaughter_i + \pi_2 \cdot (HasDaughter_i \times Male_{\mathrm{i}}) + \pi_3 \cdot Male_{\mathrm{i}} + X_i \cdot \pi + u_i$$

$Male_{\mathrm{i}}$ is a dummy variable indicating whether the respondent is a male. $HasDaughter_i$ and $HasDaughter_i \times Male_{\mathrm{i}}$ are the endogenous variables that we instrument using $FirstBornGirl_i$ and $FirstBornGirl_i \times Male_{\mathrm{i}}$. We use heteroskedasticity-robust standard errors. Other variables are defined as in H1.

Interpretation of results

The parameter of interest is π_2 which captures the additional effect on the outcome for men compared to women. We consider that the evidence is consistent with H4 if the estimate of π_2 is positive and the two-sided p-value is below 0.05.